

Modelling the hydrological response of debris-free and debris-covered glaciers to present climatic conditions in the semiarid Andes of central Chile

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Abstract. We apply the process-based, distributed TOPKAPI-ETH glacio-hydrological model to a glacierized catchment (19% glacierized) in the semiarid Andes of central Chile. The semiarid Andes provides vital freshwater resources to valleys in Chile and Argentina, but only few glacio-hydrological modelling studies have been conducted and its dominant hydrological processes remain poorly understood. The catchment contains two debris-free glaciers reaching down to 3900 m asl (Bello and Yeso glaciers) and one debris-covered avalanche-fed glacier reaching to 3200 m asl (Piramide Glacier). Our main objective is to compare the mass balance and runoff contributions of both glacier types under current climatic conditions. We use a unique dataset of field measurements collected over two ablation seasons combined with the distributed TOPKAPI-ETH model that includes physically-oriented parameterizations of snow and ice ablation, gravitational distribution of snow, snow albedo evolution and the ablation of debris-covered ice. Model outputs indicate that while the mass balance of Bello and Yeso glaciers is mostly explained by temperature gradients, the Piramide Glacier mass balance is governed by debris thickness and avalanches and has a clear non-linear profile with elevation as a result. Despite the thermal insulation effect of the debris cover, the mass balance and contribution to runoff from debris-free and debris-covered glaciers is similar in magnitude, mainly due to elevation differences. However, runoff contributions are distinct in time and seasonality with ice melt starting approximately four weeks earlier from the debris-covered glacier, what is of relevance for water resources management. At the catchment scale, snowmelt is the dominant contributor to runoff during both years. However, during the driest year of our simulations, ice melt contributes $42\pm 8\%$ and $67\pm 6\%$ of the annual and summer runoff, respectively. Sensitivity analyses show that runoff is most sensitive to temperature and precipitation gradients, melt factors and debris cover thickness.

Key words: Glacio-hydrological models, debris-covered glaciers, semiarid Andes,

TOPKAPI-ETH

1. Introduction

The cryosphere of the Andes Cordillera significantly contribute freshwater to streams flowing into the semiarid regions of central Chile and Argentina (~33.5°S). Accurate quantification of snow and ice melt is of critical importance for water managers in this region due to a heightened demand for water (Meza et al., 2015) exacerbated by a recent prolonged drought period (Boisier et al., 2015). Hydrological studies that model melt from glaciers and seasonal snow cover remain, however, scarce (Favier et al., 2009; Ragettli and Pellicciotti, 2012; Ragettli et al., 2013a) or outdated (Peña and Nazarala, 1987), and have not explicitly considered the contribution from debris-covered and rock glaciers, which are common features in the region (Azócar and Brenning, 2010; Bodin et al., 2010).

Excluding local efforts from the Chilean Water Authority (Dirección General de Aguas, DGA), which have investigated high-elevation water resources with more statistical approaches (Peña et al., 1985), the first physically-oriented modelling study to quantify runoff sources in this region was performed by Peña and Nazarala (1987). They used a semi-distributed empirical snowmelt-runoff model to simulate the hydrology of the Maipo river basin (4837 km², 8% glacierized) to the east of the Chilean capital city, Santiago. Peña and Nazarala (1987) calculated that glacier melt, i.e. melt from the ice and the snow cover over the glacier, represented up to 67% of the total runoff at the outlet of the catchment (800 m asl) during one of the driest summers on the hydro-meteorological record (1968/69). More than 25 years later, the estimates of Peña and Nazarala (1987) are still largely cited by scientists, private companies, NGOs and stake-holders (e.g. Larraín, 2007; Bown et al., 2008; Le Quesne et al., 2009; Ohlanders et al., 2013). There is a real need for updated glacio-hydrological studies due to changing climate patterns (Carrasco et al., 2005, 2008), new understanding of the main drivers of changes at high elevations and a growing demand for water resources (Meza et al., 2013). A couple of recent studies have shed light on the

hydrology of high-elevation glacierized catchments of this region (Favier et al., 2009; Ragettli and Pellicciotti, 2012; Ragettli et al., 2013a), but their dynamics remain poorly understood, key components of the cryosphere are still neglected and studies have been confined to few well monitored catchments (Pellicciotti et al., 2014).

Ragettli and Pellicciotti (2012) used the glacio-hydrological model TOPKAPI-ETH to simulate the hydrology of the Juncal Norte River catchment (241km², 14% glacierized) during two hydrological years (2005-06 and 2008-09), based on detailed hydro-glaciological measurements. They found that ice melt amounted to approximately 47% of total contributions to runoff at the catchment scale (outlet at 2233 m asl) in the 2005 late ablation season (February-April). In a following study, Ragettli et al. (2013a) compared these simulations to those of a simpler, conceptual and semi-distributed model lacking glacier representation and a snow gravitational distribution algorithm (Water Evaluation and Planning, WEAP) (Yates et al., 2005). They found that while the conceptual model can be calibrated to reproduce present-day streamflow with the same degree of accuracy as the more physically-oriented one, their future projections of runoff diverge significantly, and the inclusion of the main cryospheric processes is needed for more reliable estimates of future runoff. Ohlanders et al. (2013), estimated the water balance of the Juncal Norte River catchment using measurements of stable isotopes and found that for the unusually dry year of 2011/12 ice melt amounted to 50-90% of the total annual streamflow at the catchment outlet.

Most cryospheric studies in the semiarid Chilean Andes have looked at large scale snow (Masiokas et al., 2006, 2010; Carrasco et al., 2008), mass balance (Mernild et al., 2015) and hydro-meteorological patterns (Rubio-Álvarez and McPhee, 2010; Cortés et al., 2011; Demaria et al., 2013) or at single, specific processes, such as energy balance studies (Brock et al., 2007; Pellicciotti et al., 2008; MacDonell et al., 2013a), wind snow redistribution (Gascoin et al., 2013), or radiation fluxes over glacier surfaces (MacDonell et al., 2013b; Lhermitte et al., 2014). Most large scale studies are based on gridded data sets or remotely

sensed data (Favier et al., 2009; Mernild et al., 2015; Cornwell et al., 2016), and the detailed process-based studies are mostly observational (Gascoin et al., 2011; Sinclair and MacDonell, 2015) or use point scale models (Pellicciotti et al., 2008), so that a clear gap emerges between the two scales and type of studies, with distributed, physically-oriented, glacio-hydrological modelling studies almost entirely lacking in the region (Pellicciotti et al., 2014)

Additionally, rock and debris-covered glaciers are often excluded from hydrological analyses in the region. These features have been documented in mapping and classification studies (Bodin et al., 2010; Nicholson et al., 2010; Janke et al., 2015) but rarely modelled or included in glacio-hydrological studies. For example, Azócar and Brenning (2010) estimated that the water equivalent of ice stored in rock glaciers in the Chilean Andes between 27° and 33°S is 2.37 km³ with average thinning rates in the order of 0.6-0.7 mm yr⁻¹. In another study, using GPR observations in Tapado Glacier foreland, Pourrier et al. (2014) showed that the internal structure of rock glaciers can be highly heterogeneous with the presence of massive ice lenses. In other regions of the world, it has been shown that despite the assumed insulation effect of the debris, debris-covered glaciers might lose as much mass as debris-free glaciers (Gardelle et al., 2012, 2013; Kääb et al., 2012), mainly due to their usually lower elevation (Fujita and Sakai, 2014; Ragettli et al., 2015), presence of ice cliffs (Buri et al., 2015; Steiner et al., 2015) and supraglacial ponds (Sakai et al., 2000; Miles et al., 2016). Some of that evidence has been provided by large scale studies based on remote sensing and is still controversial (e.g. Ragettli et al., 2016). Available observations suggest however that the mass balance of debris-covered glaciers is considerably different to the one of debris-free glaciers, with highly heterogeneous surface lowering (Immerzeel et al., 2014; Kraaijenbrink et al., 2016; Ragettli et al., 2016) and non-linear mass balance gradients (Nuimura et al., 2012; Pellicciotti et al., 2015). To our knowledge, no study has supported this evidence with results from a physically-oriented model that explicitly incorporates the processes that are believed to be dominant on debris-covered glaciers.

In this paper, we build on the work of Ragettli and Pellicciotti (2012) and Ragettli et al. (2013a) to study the hydrological contribution of debris-covered and debris-free glaciers. Our main aims are to: i) compare the mass balance and hydrological contribution of these two glacier types and ii) quantify the magnitude and timing of the contribution of snow and ice melt to total runoff at the catchment scale.

2. Study catchment

The Río del Yeso catchment is located in the semiarid Andes of central Chile, 70 km east of Santiago (Figure 1). The catchment has an area of 62km² (19% glacierized), a mean elevation of 4007 m above sea level (asl) and it provides freshwater resources to El Yeso reservoir, which is used to manage the drinking water supply for Santiago. In this study we focus on the three major glaciers of the catchment: Bello, Yeso and Piramide (Table 1).

Despite their key location, very little research has been conducted at Bello and Yeso glaciers. To our knowledge, the only available literature comes from DGA technical reports and inventories, which are mostly descriptive (Marangunic, 1979). Bello Glacier has a south-east aspect and a southward flowing tongue, partially debris-covered at its terminus. Yeso Glacier is located next to Bello Glacier, east of a ridge descending from Bello peak (5230 m asl). It has a south-west aspect and its tongue descends in a south-west direction. Both glaciers contribute to the Yeso River, which extends to the south for approximately 8 km before reaching the outlet defined for this study.

Piramide Glacier is a debris-covered glacier located in the neighbouring valley to the east of Yeso River. The glacier has no classically-shaped accumulation area and is mainly fed by avalanches generated from the steep slopes surrounding its upper area. The glacier extends southward for about 6 km before reaching its terminus, about 2 km away from the Río del Yeso catchment outlet. The glacier is stagnant with advance velocities close to zero, a fact that can be corroborated by the absence of ridges and furrows, which are typically found on active debris-covered glaciers. Ferrando (2012) suggests that its chaotic surface topography is a product of differential ablation. Piramide Glacier has experienced a small

area loss compared to other glaciers in this region (Janke et al., 2015), which is typical of debris-covered glaciers, which do not recede but rather thin or downwaste. According to Janke et al. (2015), the glacier has not yet reached a sufficient debris cover thickness to enable the thermal insulation of the ice body.

3. Data description

3.1 Hydro-meteorological data

An automatic weather station (AWS) was installed on each of the three glaciers (Figure 1 and Table 2). Due to logistical constraints, e.g. sites with a difficult access, or locations too close to potentially dangerous slopes during winter months, the AWSs were installed at different times and some were moved during the course of the study. Another AWS was located off-glacier on the slopes surrounding Yeso and Piramide glaciers. This station was positioned near the ablation tongue of Yeso Glacier in 2013-14 and moved to a location next to Piramide Glacier in 2014-15. All AWSs recorded data every hour, with the exception of the AWS on Piramide Glacier that recorded data every ten minutes. For this study all data were aggregated to hourly averages. A summary of the AWS locations, instruments and periods of functioning is shown in Figure 1 and Table 2.

In addition to the AWSs data, we used data from the DGA meteorological network in the lower valleys to estimate off-glacier gradients of precipitation and air temperature (Table 3). In order to derive cloud cover transmittance, we used the daily incident radiation ($1^\circ \times 1^\circ$ gridded product) from the “Climatology Resource for Agroclimatology” project in the NASA Prediction Worldwide Energy Resource “POWER” (<http://power.larc.nasa.gov/>). Finally, discharge was manually measured using salt dilution gauging at the outlet of Bello and Yeso glaciers during the ablation season (Figure 1).

3.2 Glaciological measurements

Measurements of snow depth and surface lowering were performed at Bello and Piramide glaciers in 2013 and 2014 (Figure 1). On Bello Glacier, nine ablation stakes were

installed on 16 January 2014 along the glacier centerline. On Piramide Glacier, nine ablation stakes were first installed on snow on 6 November 2013 and replaced by a network installed into debris and ice once the snowpack disappeared on 29 January 2014. This second network was maintained for the rest of the study period. Snow depth measurements were performed at the end of the accumulation season on 5 November 2013 (Piramide Glacier) and on 1 October 2014 (Piramide and Bello glaciers). The location of these measurements was chosen to cover the maximum range of elevation along the centre flow line of each glacier. At each point several snow depth measurements were performed and averaged.

Two snowpits were dug next to the AWS on Piramide Glacier to measure snow density and calculate snow water equivalent. The first snowpit was dug on 5 November 2013 (2.20 m depth, 520 kg m^{-3} mean density), and the second on 1 October 2014 (0.7 m depth, 280 kg m^{-3} mean density). One snowpit was dug next to the AWS on Bello Glacier on 1 October 2014 (1.7m depth, 310 kg m^{-3} mean density). Finally, debris cover thickness was manually measured at each ablation stake on 29 January 2014 on Piramide Glacier. These data were extended using seven additional measurements performed during 2012 (Comitato-Ev-K2-CNR, 2012).

3.3 Terrestrial photos

A time-lapse camera (Harbortronics system) was installed in front of Bello Glacier on 27 February 2014 to take daily photos of the lower section of the glacier (Figure 1). The time-lapse system included a Canon EOS Rebel T3 camera with a resolution of 12.2 MP and a focal length of 18 mm and was programmed to take photos at 13h (Chilean summer time). We obtained a total of 48 valid photos in the period 27 February 2014 to 15 April 2014 and 114 valid photos in the period 23 October 2014 to 1 April 2015. The extension of the monitored area is 0.84 km^2 , which corresponds to 20.5% of the total glacier surface of Bello Glacier. The daily photos were used to calculate distributed maps of surface albedo and to derive snow-covered areas (Section 5.2).

3.4 Remote sensing products

The topography of the catchment was obtained from a Digital Elevation Model (DEM) of 30 m resolution, acquired on February 2000, extracted from the Shuttle Radar Topographic Mission (SRTM). Additionally, we used a LANDSAT 8 satellite thermal image of 60 m resolution retrieved on 22 February 2015 to derive surface temperature of debris-covered areas. Finally, glacier outlines were extracted from the Chilean National glacier inventory (DGA, 2010).

4. The TOPKAPI-ETH model

4.1 Model description

The TOPKAPI-ETH model is a fully distributed process-based hydrological model, which was first developed by Ciarapica and Todini (2002) and Liu and Todini (2002) as a rainfall-runoff model. The main characteristic of the original model is the use of the kinematic approximation to route subsurface, overland and channel flow. Further developments have provided a set of new modules that make the model suitable for water budget analyses of high-elevation areas dominated by cryospheric processes (Finger et al., 2011; Ragettli and Pellicciotti, 2012; Ragettli et al., 2013a, 2013b, 2015; Fatichi et al., 2015). These modules include snow and ice melt (Pellicciotti et al., 2005), melt of debris-covered ice (Carenzo et al., 2016), a parameterisation of glacier geometry changes implicitly accounting for glacier flow (Huss et al., 2010) snow albedo evolution (Brock et al., 2000), redistribution of snow by avalanching (Bernhardt and Schulz, 2010) and melt water routing (Hock and Noetzli, 1997).

Melt from snow and bare-ice is computed in each grid cell using the Enhanced Temperature-Index (ETI) model (Pellicciotti et al., 2005):

$$M = \begin{cases} SRF \cdot I \cdot (1 - \alpha) + TF \cdot T & T > T_T \\ 0 & T \leq T_T \end{cases} \quad (1)$$

, where M [mm h^{-1}] is hourly melt; SRF [$\text{m}^2 \text{ mm W}^{-2} \text{ h}^{-1}$] and TF [$\text{mm hr}^{-1} \text{ }^\circ\text{C}^{-1}$] are shortwave radiation and temperature factors, respectively; I [W m^{-2}] is the incoming shortwave

radiation; α [-] is the surface albedo; T [°C] is the air temperature and T_T [°C] is an air temperature threshold for the onset of melt.

TOPKAPI-ETH computes melt under debris using a modification of the ETI model, which is referred to as D-ETI (Ragetti et al., 2015; Carenzo et al., 2016). In this approach the coefficients SRF and TF are parameterised as a function of debris thickness. A lag parameter, which is also a function of debris thickness, is introduced to represent the time needed for the energy transfer through the debris layer. The resulting set of equations is:

$$M = \begin{cases} SRF_d \cdot I(t - lag) \cdot (1 - \alpha) + TF_d \cdot T(t - lag) & T \geq T_T \\ 0 & T < T_T \end{cases} \quad (2)$$

$$SRF_d = SRF_{d1} \cdot e^{-SRF_{d2} \cdot d} \quad (3)$$

$$TF_d = TF_{d1} \cdot d^{-TF_{d2}} \quad (4)$$

$$lag = lag_{d1} \cdot d - lag_{d2} \quad (5)$$

, where d [m] is debris thickness and SRF_{d1} [mm m² hr⁻¹ W⁻¹], SRF_{d2} [m⁻¹], TF_{d1} [mm hr⁻¹ °C⁻¹], TF_{d2} [-], lag_{d1} [hr m⁻¹] and lag_{d2} [hr] are parameters that were calibrated against the results of an energy balance model as a reference.

The evolution of snow albedo is calculated using the approach by Brock et al. (2000):

$$\alpha = \alpha_{max} - r_d \cdot \log_{10} T_{acc} \quad (6)$$

, where α_{max} [-] is the albedo of fresh snow, r_d [-] is a decay rate and T_{acc} [°C] is the accumulated daily positive maximum air temperature.

Snow gravitational redistribution is implemented using the SnowSlide model of Bernhardt and Shulz (2010), in which snow available for transport corresponds to the difference between the grid cell snow holding depth (S_{hd} [m]) and the current snow depth. The value of S_{hd} is calculated as a function of the slope angle:

$$S_{hd} = SGR_C \cdot e^{SGR_a \cdot SLP} \quad (7)$$

, where SGR_C and SGR_a are empirical parameters and SLP is the grid slope.

Snow is transferred to the lower neighbouring cells and is portioned based on the vertical distance between neighbouring and initial grid elements. The scheme allows for divergent and convergent flow and is strictly mass conserving. TOPKAPI-ETH simulates discharge from glaciers using two linear reservoirs that represent the storage of meltwater within the snowpack (k_{snow} [h]) and ice (k_{ice} [h]) (Hock and Noetzli, 1997). The long term evolution of glacier geometry in response to climate forcing is also included in the model through a conceptual parameterization of ice flow dynamics (Huss et al., 2010; Ragettli, 2014). However, the time scale of these changes is typically in the order of several years or decades, depending on glacier scale. Due to the short time span of our simulations, the ice flow module was switched off, as no significant geometry changes were expected to take place over the two years of analysis.

4.2 Forcing variables

TOPKAPI-ETH requires hourly time series of air temperature, precipitation, clear-sky incoming shortwave radiation and daily cloud transmittance factors. Air temperature and precipitation time series from stations of the DGA meteorological network were distributed over the catchment. Air temperature was distributed using monthly mean lapse rates (T_{grad}) (Table 4). To represent the temperature variations induced by the glacier boundary layer (Greuell and Böhm, 1998; Brock et al., 2010), we used two parameters to decrease air temperature on debris-free glaciers (T_{mod}) and increase it on debris-covered glaciers ($T_{mod_{debris}}$):

$$T(z) = \begin{cases} T(z_0) + T_{grad} \cdot (z - z_0) - T_{mod} & \text{Debris - free cell} \\ T(z_0) + T_{grad} \cdot (z - z_0) + T_{mod_{debris}} & \text{Debris - covered cell} \end{cases} \quad (8)$$

,where z is the elevation of each grid cell and z_0 is the elevation of the base station. We distributed precipitation using logarithmic altitudinal gradients:.

$$P(z) = [P_{grad_c1} \cdot \ln(z) + P_{grad_c2}] \cdot P(z_0) \quad (9)$$

, where $P(z)$ [mm] is precipitation at elevation z [m asl], $P(z_0)$ [mm] is precipitation recorded at the base station and P_{grad_c1} and P_{grad_c2} are calibrated parameters.

Clear-sky incoming shortwave radiation for each grid cell is calculated by TOPKAPI-ETH following the approach described in Pellicciotti et al. (2011). Finally, the ratio between the daily incident radiation ($1^\circ \times 1^\circ$, NASA product) and the potential radiation was used to compute cloud transmittance factors, which were considered uniform for the entire catchment.

4.3 Calibration and validation strategy

The period of analysis of our study is 1 April 2012 to 31 March 2015. This period includes the two monitored years (2013-14 and 2014-15) plus the year 2012-13, which was simulated in order to reach a model state independent from arbitrarily selected initial conditions. Due to the limited extension of our dataset and our focus on reproducing processes in the best possible manner we did not use a traditional division between calibration and validation periods and instead we defined a strategy based on the work of Ragetti and Pellicciotti (2012) and Ragetti et al. (2015) (Figure 2). In this way, field measurements that represent a single hydrological process were used to calibrate the parameters controlling that process and were not further modified during the calibration of parameters that resulted from the integration of several processes at the glacier or catchment scale. For example, the parameters T_{mod} and T_{mod_debris} were calibrated exclusively using mean temperature differences between on and off glacier AWSs. On the other hand, field measurements that result from the interaction of several processes were mostly used as validation datasets. For example, time series of distributed albedo measurements were used for validation because they depend on snow accumulation, melt

processes and albedo decay. When we calibrated a parameter that integrates the influence of many other processes we did not modify the previously calibrated parameters. In contrast to common hydrological practice, and also considering our limited amount of streamflow data, we did not use runoff to calibrate model parameters related to distribution of meteorological variables or melt processes. In this way, we can minimise equifinality problems that typically arise in hydrological modelling from the use of single-variable calibration schemes (Beven and Binley, 1992; Beven, 2006; Kirchner, 2006).

The detailed conceptual scheme of this procedure is shown in Figure 2. In step 1 we calculated monthly average temperature gradients using the off-glacier DGA weather network and we calibrated T_{mod} and $T_{mod_{debris}}$ using the average differences between on-glacier and off-glacier AWSs. In step 2 we derived the ETI and D-ETI parameters using as reference hourly melt rates calculated with a point-scale energy balance model at the AWSs on Bello and Piramide glaciers. The energy balance model at Bello AWS was ran for the period 15 November 2013 to 20 January 2014 and at Piramide AWS for the period March 2015, when snow was absent. The parameters controlling the albedo of snow covered grid cells (α_{max} and r_d) were calibrated using albedo data from the AWS on Bello Glacier in 2014-15, as this period had a higher variability in albedo values. In step 3 we calibrated precipitation gradients using DGA meteorological stations and the manual snow depth measurements. In step 4 we validated the results from the first three steps using hourly temperature measurements at each AWS, the ablation stakes network and time series of distributed albedo and snow cover area derived from the time-lapse camera. Finally, in step 5, we calibrated the storage coefficients of the snow and ice surfaces using streamflow data at the outlets of Bello and Yeso glaciers.

5. Additional methods

5.1 Debris cover thickness estimation

We derived a debris thickness map (Figure 3) using the methodology proposed by Rounce and McKinney (2014). In this method, a cloud-free thermal satellite image is used to estimate distributed surface temperature, which is then used to solve the surface energy balance in every grid cell of the glacier area. A detailed description of this method is provided in Appendix A1.

Bello and Yeso glaciers show patches of thin debris (<5 cm) with some areas of thick debris at the glacier terminus (>20 cm). Piramide Glacier is completely debris covered with some areas of thin debris (<5 cm) in the upper section, but more than half of the glacier is covered by a debris layer thicker than 10 cm. We compared these results to the dataset of point debris thickness measurements (Figure 3) and found that the observations are well reproduced for debris thickness thinner than 20 cm and the results largely capture observed spatial variability. The agreement for greater thickness is not good. This might be due to both the model failing as with greater sediment thickness, surface temperature becomes less dependent on sediment thickness (Schauwecker et al., 2015) and to errors in the observations for thicker measurements.

5.2 Reconstruction of albedo from terrestrial photos

We used the method by Corripio (2004) to estimate distributed albedo on Bello Glacier from the daily photographs taken by the time-lapse camera. All cloud-free photographs were georeferenced using the camera position and settings in conjunction with the DEM of the area. Reflectance values were converted to albedo using point albedo measurements at the AWS, atmospheric transmissivity (calculated using the software MODTRAN (Berk et al., 2008)), estimations of diffuse and direct irradiation and the sky view factor of each pixel. Additionally, we estimated the snow covered area of the monitored

surface assuming a threshold of 0.3 in the albedo values to distinguish between snow-free and snow-covered areas.

5.3 Sensitivity and uncertainty analysis

To evaluate the accuracy of model results, we performed a sensitivity analysis of model outputs to parameter variations, as well as an uncertainty analysis of model results. We calculated the sensitivity of total annual runoff to parameter variations for the 2014-15 hydrological year following the approach of Anslow et al. (2008) as used by Ragettli and Pellicciotti (2012) and Heynen et al. (2013). This analysis corresponds to a “one-at-a-time” approach, in which sensitivity coefficients were calculated by varying each parameter by a given amount, calculating the corresponding change in runoff and then fitting a second order polynomial to the obtained set of values (for details of the curve fitting see Figure 4 in Ragettli and Pellicciotti (2012)). The sensitivity coefficient is the slope of the tangent to the polynomial in correspondence to the initial set. In this way, we calculated the rate of change of the output variable to variations in the individual parameter values and identified the most sensitive parameters. Additionally, for each parameter we selected the values for which the total annual runoff varies by less than $\pm 5\%$, to provide a range of plausible parameters that would result in an acceptable model performance.

After identifying the most sensitive model parameters, we performed an uncertainty analysis for the 2014-15 period. We conducted 250 additional simulations, with each parameter set containing randomly selected parameters within the obtained intervals of acceptable model performance. The parameter selection was performed using Sobol' quasi-random number generator to efficiently sample the parameter space (Bratley and Fox, 1988). Uncertainty ranges of the model outputs are provided using the median and the standard deviation of the 250 additional simulations.

6. Results

6.1 Calibration and validation

Overall model calibration provided a good agreement between modelled and measured quantities, and so the calibrated parameters can be assumed to be adequate for use within the modelled basin over the study period (Table 4). The results of each calibration step are outlined below.

The calibration of air temperature distribution (Step 1, Figure 2) showed that air temperature lapse rates are steeper during summer (Table 4), which is consistent with previous studies in alpine regions (Rolland, 2003; Blandford et al., 2008). Calibration of the ETI and D-ETI parameters (Step 2, Figure 2) shows a relatively good agreement between reference and simulated melt rates (Nash-Sutcliffe >0.88 , Figure 4; Table 4). Figure 5a shows the results of the calibration of precipitation gradients using the snow depth measurements (Step 3 of Figure 2). Snow depth measurements at Bello and Piramide glaciers can only be reproduced by using negative relationships with the elevation (Table 4). Precipitation at the outlet of the catchment was approximately three times higher than that at the base station, a value that is consistent with the literature (Favier et al., 2009; Ragettli et al., 2013a), resulting in an increasing trend below 3000 m asl and a decreasing trend above this elevation.

Results from the model validation (Step 4, Figure 2) indicate a relatively good agreement between observed and simulated hourly values of air temperature over the studied glaciers. Simulated air temperatures on Piramide Glacier show the best agreement with observations ($R^2=0.92$), suggesting that variations of air temperature induced by the debris-covered glacier boundary layer are relatively small in comparison to the ones induced by the boundary layer over debris-free glaciers (as described in Brock et al. (2010)). Simulations of air temperature at Bello and Yeso glaciers show a poorer agreement (R^2 values of 0.5 and 0.76, respectively), particularly at the high-elevation location of Yeso AWS (4428 m asl), which is probably due to complex interactions of synoptic systems and local

glacier winds that undermine the assumption of uniform lapse rates (Shea and Moore, 2010; Petersen et al., 2013; Ayala et al., 2015). In the second validation step, we compared ablation at the ablation stakes and modelled by TOPKAPI-ETH (Figure 5b). While the medians of the observations are well reproduced by the model, the spatial variability is not always captured. This can be related to errors in the estimated debris cover thickness map, the simulation of the end-of-winter snow accumulation or differences between the local and the 30 m scale of the DEM. The final validation step was performed using distributed albedo maps derived from the terrestrial time-lapse camera (Figure 6). There is an overall good correspondence between observed and simulated values for year 2014-15, but results were poorer for year 2013-14 when fewer photographs were available (not shown). Many reasons could explain the differences in observed and simulated albedo, including precipitation gradients not appropriate for specific events, air temperature gradients not appropriate at that time and/or parameters controlling melt and albedo decay that were not valid for the specific meteorological conditions.

Finally for the calibration of glacier runoff (Step 5, Figure 2), we found a good agreement between observed and simulated values of streamflow, but there is probably uncertainty in these parameters, because the data used for calibration are scarce (Figure 7).

6.2 Glacier mass balance

Figure 8 shows the annual glacier mass balance of each year as a function of elevation and debris-free and debris-covered areas. In 2013-14, 51% more precipitation was registered at El Yeso reservoir than in 2014-15. As a result, TOPKAPI-ETH shows a less negative mass balance during that season.

The mass balance patterns of Bello (Figure 8a) and Yeso (Figure 8b) are remarkably distinct from that of Piramide (Figure 9c). While the mass balance of the former has a clear relation with altitude, with an approximate linear trend above the debris-covered patches at the terminus, the mass balance profile of the debris-covered Piramide Glacier is non-linear, with less negative mass balance at the lower elevations and is increasingly negative with

higher elevation, where debris becomes thinner favouring higher melt rates (from 3200 to 3800 m asl). In the upper reaches (from approximately 4100 m asl), where the debris cover is thinner, the usual profile is re-established, such that the mass balance is less negative at higher elevations following an approximately linear behaviour.

To quantify the effect of snow avalanches on the glacier mass balance, we ran a simulation of the model with the snow gravitational distribution module switched off. Figure 8 shows that snow gravitational redistribution can change the local mass balance considerably, turning it from negative to positive, especially in the upper part of Piramide between 3800 and 4200 m asl. Above this glacier section, we observe the opposite effect; high-elevation steep areas lose mass. The effect of avalanches is more pronounced in 2013-14 than in 2014-15 because of the higher precipitation amount received during the former period. Mass balance is considerably more negative when avalanches are not included due to the contribution of snow from non-glacierized steep areas (Figure 8d), especially in the higher precipitation year 2013-14. The effect on the mass balance profile is important for Piramide Glacier in particular, because of its lower elevation where temperatures are higher.

In Figure 8d and Table 5, we show a summary of total glacier mass balance and the equilibrium line altitude (ELA) for both years with and without avalanches. The glaciers have mostly negative annual mass balances, however Bello and Yeso glaciers were close to a neutral mass balance in 2013-14. ELAs are between 4400 and 4900 m asl for these debris-free glaciers. These values define an accumulation area between one half and one third of the total glacier size. As the mass balance of a debris-covered glacier is not controlled by elevation, it is difficult to define the ELA for Piramide Glacier. We found values between 3800 and 4100 m asl, but these values are strongly influenced by the mass gravitational redistribution.

6.3 Streamflow and runoff components

Figure 9 (a-d) shows simulations of daily discharge at the outlet of Bello, Yeso and Piramide glaciers from the ice and snow reservoirs of TOPKAPI-ETH. Differences between

the two years are evident: while snowmelt dominates discharge in 2013-14 (Figure 9a and 9b), ice melt is more important in 2014-15 (Figure 9c and 9d). Due to its lower elevation, snow and ice melt from Piramide Glacier starts one or two months earlier than in Bello and Yeso glaciers. By the end of summer, the contribution of ice melt to discharge at Piramide Glacier is of comparable magnitude to the one of Bello and Yeso glaciers, which are located at considerable higher elevations, but with ice surfaces directly exposed to the atmosphere. Annual averages of runoff are shown in Figure 9e. Piramide Glacier contributes the most to runoff, particularly in 2013-14 (Figure 9a). However, in the drier year 2014-15, the contribution of Bello and Yeso glaciers increases. When we plot the specific runoff generated from each glacier (Figure 9b), it becomes clear that all glaciers contribute in a similar magnitude.

Figure 10 shows the contribution of snowmelt (on and off glacier), glacier ice and liquid precipitation to catchment runoff. Snow cover is the main contributor to runoff in both years, but ice melt can increase from 15% in a snow-rich year such as 2013-14 to 35% in 2014-15 and to more than 50% during the austral summer (Figure 10b). The contribution of rain is negligible in both years.

6.4 Sensitivity and uncertainty analyses

The sensitivity analysis showed that the model is most sensitive to T_{grad} , SRF , T_T , α_{max} , $P_{grad\ C1}$ and $P_{grad\ C2}$ (Table 4). Figure 11 shows uncertainty ranges for the results of Figures 8d, 9e, 9f and 10b for year 2014-15. The differences in the normalized runoff contribution (Figure 11b) and mass balances (Figure 11c) of the three glaciers are not significant when uncertainty estimates are included. Figure 11d shows that the ice melt contribution to the catchment runoff is $42\pm 8\%$ during the entire year and $67\pm 6\%$ during the summer season (January to March).

7. Discussion

7.1 Mass balance and runoff results

The mass balance pattern of Bello and Yeso glaciers, with an approximate linear relation with elevation, agrees with findings from debris-free glaciers in other regions of the world (Vincent, 2002; Huss et al., 2008; Wagnon et al., 2013). We found that ELAs in the debris-free glaciers are between 4400 and 4900 m asl, some of which are higher than what Ragettli et al. (2013a) obtained for Juncal Norte Glacier at the end of the 2009 summer (4500 m asl) and that which Carrasco et al. (2005) estimated for the region using radiosonde data at the end of the 20th century (4200 m asl). This might be due to either differences in glacier characteristics and elevation in particular, or to the retreat of the ELA to higher altitudes in the most recent period (Mernild et al., 2015). The ELA in Piramide Glacier is considerably lower (between 3800 and 4100 m asl), which is consistent with results from other debris-covered glaciers (Ragettli et al., 2015), but it is difficult to properly define an ELA when the elevation is not the main control on the mass balance. In a recent study, Mernild et al. (2015) found that during the period 2003-2012 the average annual mass balance of glaciers in the central Andes was about -0.86 ± 0.25 m water equivalent (we). This value is close to the average mass balance of the two years analysed in this study, which is -1.0 ± 0.11 m.

Avalanches play a key role for all glaciers but especially at Piramide Glacier, where the low-lying tongue is fed by large avalanches from the steep walls surrounding the upper glacier (Ferrando, 2012). Debris-covered glaciers in many regions of the world are avalanche-fed, but the actual contributions of avalanches to total mass balance and melt have been rarely quantified (Ragettli et al., 2015) and this is the first study that shows this effect at play for the Andes of Chile. Our results show that neglecting the avalanche-redistributed snow can affect modelled mass balance results considerably, as avalanches can modify the sign of local mass balances. Simulation of avalanches thus seems an

important component for glacio-hydrological models applied to the high relief glacierized catchments of the Andes of Chile.

It is clear from our reference simulation that snowmelt is, at the scale of the catchment investigated, the most important contributor to total runoff, with 84% and 66 % in 2013-14 and 2014-15, respectively. Nevertheless, ice melt is an important source of water in dry years (30% in 2014-2015) and summer. These results for the ice melt contribution are higher than those of Ragettli and Pellicciotti (2012) using the same model for year 2005-06 in Juncal Norte River catchment (14% of total annual runoff), but the contribution of ice melt is lower than the one calculated by Ohlanders et al. (2013) for the unusually dry year of 2011-12 (50-90% of total annual runoff). The contribution to runoff of ice melt during summer is about 54% in 2014-15, which is lower than the numbers found by Peña and Nazarala (1987) for summer 1968-69 at a much lower elevation. However, it is difficult to establish whether this is due to differences in the methodology or the climatic conditions of those periods or a combination of both. We have also demonstrated that glacier melt depends on the characteristics of the glaciers, elevation range and surface features, so that any generalisations should be made with caution. Instead, it seems important to repeat such integrated data-rich and modelling studies for other catchments with similar availability of ground observations to establish regional patterns of glacier and runoff changes.

7.2 Sources of uncertainty

TOPKAPI-ETH parameters have been calibrated using extensive, local, in-situ collected data sets to avoid error compensation and parameter ambiguity, a problem that arises when numerous parameters are calibrated only against one or two integrated catchment-response variables, such as runoff and/or snow cover (Finger et al., 2011; Pellicciotti et al., 2012). The good agreement between measured and calculated quantities indicate that the model can accurately reproduce several key processes that contribute to the hydrological response of the catchment and is therefore appropriate to characterise Andean catchments. Nevertheless, there remain three main sources of uncertainty in our

study: the spatial distribution of forcing variables, the debris thickness reconstruction and the seasonal variability of the calibrated parameters.

While the altitudinal gradients might be able to reproduce precipitation at large, regional scales, they are likely unable to reproduce the complex spatial patterns on rough, high-elevation catchments (Molotch et al., 2005; Lehning et al., 2011). Our observations of snow water equivalent were particularly low at high elevations, suggesting the action of snow removal by wind, more local complex patterns of precipitation deposition or sublimation losses. A more in-depth understanding of precipitation variability at high elevation would reduce uncertainty in hydrological simulations.

The calculated mass balance and runoff of Piramide Glacier depends primarily on the accuracy of the derived debris thickness map. The major strength of the method used in this study to derive the debris thickness is the use of physically-based equations and, in our application, the use of in-situ data. However, temperature profiles within the debris, the rate of heat storage and turbulent heat fluxes can create large uncertainties, which significantly increase with a thicker debris cover. While estimated thicknesses at Piramide Glacier are plausible and within the observed range, they might be subjected to uncertainties and values might be too thin in the upper sections (Schauwecker et al., 2015). For these reasons we tested the sensitivity of our results to changes in the debris thickness, finding that variations of $\pm 20\%$ and ± 10 cm produce changes of about $\pm 10\%$ and $\pm 50\%$ in the simulated ice melt of the catchment (with thicker debris producing less melt). These results confirm that the model is highly sensitive to debris thickness and the total runoff from Piramide Glacier might be subject to large uncertainties. However, we have confidence in our estimates due to the good model performance in the validation at ablation stakes along the entire length of the glacier.

The adopted stepwise calibration approach attempts to minimize the uncertainty in model parameters by calibrating them against direct measurements of glacio-hydrological processes. However, a limitation of our calibration strategy is that parameters were held

constant over the calibration period. This guarantees that they are representative of average conditions over the period of investigation, but also results in a loss of information content. During periods when the snowpack temperature drops below 0°C or processes other than melt become important to the mass balance (such as snow sublimation) the performance of the ETI model decreases if parameters are not recalibrated for those conditions (Ayala et al., 2016). Indeed, the ETI model overestimates daily melt peaks during the first half of the simulated period and compensates by underestimating daily peaks in midsummer (Figure 5a).

In addition to these three sources of uncertainty, there are two processes that were not explicitly considered by the model, which can be relevant for this region. These processes are snow sublimation and the hydrological significance of rock glaciers. Snow sublimation amounts for approximately 20% of total ablation below 5000 m asl (Pellicciotti et al., 2008; Ayala et al., 2016), and should therefore not strongly affect the simulations in this study (50% and 83% of the catchment is below 4000 and 4500 m asl, respectively). However, the errors induced by neglecting snow sublimation may be compensated by other parameters in the model and snow sublimation amounts could have a relevant role for the highest sites and for the long-term modelling of glacier mass balance.

Due to a basic lack of understanding, a parameterization of the hydrological contribution of rock glaciers has not been yet explicitly incorporated into glacio-hydrological models. Additionally, the method of Rounce and McKinney (2014) to estimate debris thickness was not developed for rock glaciers. However, it should be noted that rock glaciers in the catchment are small (less than 7.9% of total glacierized area) and are located above 3600 m asl. Future studies should aim to understand the hydrological behaviour of these features and provide numerical models, especially for long term simulations, but in the absence of an explicit parameterization the treatment of rock glaciers as debris-covered glaciers might give a reasonable approximation of their contribution to runoff.

8. Conclusions

We have presented the first modelling comparison of the response to present climate of debris-covered and debris-free glaciers in the Andes of central Chile. We used for this a physically-oriented and distributed glacio-hydrological model which was set, calibrated and validated using an extensive data set of in-situ observations. To our knowledge, this is one of the few existing modelling studies in the central Andes of Chile, and the first that takes into account explicitly the role of debris-covered glaciers. Our main conclusions are as follows:

- a. Debris-covered and debris-free glaciers exhibit remarkably different spatial patterns of mass balance. A positive altitudinal gradient of mass balance is evident for the debris-free Bello and Yeso glaciers, which suggests a connection to air temperature lapse rates. On the other hand, the debris-covered Piramide Glacier mass balance shows, overall, no variation with altitude. On this glacier, piecewise altitudinal gradients, of negative and positive sign, are controlled by debris cover thickness and avalanching. This result has been observed in other studies based on remote sensing data, but it has never been shown as the result of a process-based model.
- b. Despite the different spatial patterns of mass balance, debris-free and debris-covered glaciers have similar magnitudes of total mass balance and runoff contribution. However, they differ remarkably in the timing of runoff contribution as snow and ice melt from the debris-covered Piramide Glacier start approximately four weeks earlier than those of the debris-free glaciers. This result is relevant for water managers interested in the timing and seasonality of freshwater resources.
- c. At the catchment scale, snowmelt is the dominant contributor to runoff during both years. However, during the driest year of our simulations, ice melt contributes $42\pm 8\%$ and $67\pm 6\%$ of the annual and summer runoff, respectively.

- d. A sensitivity analysis showed that some of the largest uncertainties in our results arise from the distribution of the meteorological forcing and the estimation of the debris thickness. However, the sensitivity of mass balances and runoff contributions from Piramide Glacier to variations in meteorological forcing is considerably lower than the one of debris-free glaciers, confirming previous results that suggested that debris-covered glaciers react more slowly to a changing climate.

The data collection effort performed for this study resulted in a unique dataset that allowed the calibration of a process-based hydrological model, its internal validation in a stepwise manner, and a consistent comparative study of three glaciers in the catchment. Given the abundance of debris-covered glaciers in the central Andes, it seems imperative to invest into increasing our understanding of their hydrology, surface features and melt mechanisms. We strongly encourage more studies that combine collection of in-situ, detailed datasets with distributed, physically-oriented models, to establish in a consistent manner the hydrological response to the current climate of high-elevation Andean catchment as a basis for robust future projections of changes.

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Appendix A1. Calculation of debris thickness

Background

Debris thickness was computed using the method developed by Rounce and McKinney (2014), thereafter called the R14 method, which is an improved version of a methodology proposed by Foster et al. (2012), here named the F12 method. The basic principle of both methods is that debris surface temperature during daytime is largely dependent on debris thickness. The influence from other sources, such as shading, daytime and air temperature can be removed by accounting for those fluxes and solving the surface energy balance equations for a layer of debris.

In the original model by Foster et al. (2012), the ground heat flux G , increased by a stored heat factor F , is the residual of the energy balance:

$$S + L + H + LE = G \cdot (1 + F) \quad (\text{A.1})$$

, where S is net shortwave radiation, L is net longwave radiation, LE is the latent heat flux and H is the sensible heat flux.

According to Fick's law of heat diffusion, G can also be expressed as:

$$G = K \cdot \frac{dT}{dz} \simeq K \cdot \frac{T_s - T_{ice}}{d} \quad (\text{A.2})$$

, where K is the effective thermal conductivity, T_s is debris surface temperature, T_{ice} is the temperature of the underlying glacier ice ($T_{ice} = 0^\circ\text{C}$) and d is debris thickness.

Originally, the F12 method accounts for the effect of heat storage within the debris by introducing a constant heat storage factor $F=0.64$ and by assuming a linear debris temperature profile. The R14 method introduces the G_{ratio} to correct the simplified assumption of a linear debris temperature profile in the calculation of the ground heat flux G . During the warming of the debris surface by shortwave radiation in the morning hours, the temperature gradient in the uppermost part of the debris layer is very steep, while the

gradient remains small or even close to zero in the lower part. By measuring vertical profiles of debris temperature, the G_{ratio} can be computed as the ratio between the debris temperature gradient in the uppermost layer and the debris temperature gradient over the entire debris layer:

$$G_{ratio} = \frac{T_S - T_1}{d_1} / \frac{T_S - T_{ice}}{d} \quad (A.3)$$

, where d_1 and T_1 are the thickness and the temperature of the uppermost layer. The corrected ground heat flux G becomes:

$$G = G_{ratio} \cdot K \cdot \frac{T_S - T_{ice}}{d} = G_{ratio} \cdot K \cdot \frac{T_S}{d} \quad (A.4)$$

And the calculation of debris thickness from the inversion of the energy balance is:

$$d = K \cdot T_S \cdot \frac{G_{ratio}}{LE + H + (1 - \alpha)SW_{in} + L} \quad (A.5)$$

Thereby, the stored heat factor $F = 0.64$, used in the F12 method, is not needed in this formulation, because the G_{ratio} itself accounts for effects of heat storage.

Data and methodology

In our study, meteorological data necessary for the energy balance calculations are measured at the AWS on Piramide Glacier. The distributed debris surface temperature is obtained from a LANDSAT 8 satellite thermal image of 60 m resolution retrieved on 22 February 2015. The meteorological data needed as input to the energy balance equation and measured at the location of the AWS need to be distributed to the glacier scale and every pixel. For each satellite image pixel, air temperature was calculated from satellite-borne surface temperature via a partial linear regression, as described by Steiner and Pellicciotti (2016). This allows a much more accurate calculation of the turbulent sensible

heat flux and the incoming longwave radiation than if uniform in space air temperature were used as forcing to the model (Equation A.4). Distributed incoming shortwave radiation was computed using the solar radiation module of ArcGIS® software for the exact time and date of the satellite overpass. Incoming longwave radiation was computed using a parameterization by Dilley and O'Brien (1998), with parameters optimized by Juszak and Pellicciotti (2013).

As there were no specific measurements in the Río del Yeso catchment we adopted a value of $0.94 \text{ Wm}^{-1}\text{K}^{-1}$ for thermal effective conductivity (Brock et al., 2010) and a value of 0.95 for emissivity (Brock et al., 2010; Lejeune et al., 2013; Rounce and McKinney, 2014; Evatt et al., 2015). Reported values for the roughness length for momentum (z_0) for debris covered glaciers range between $z_0 = 0.0063$ (Khumbu Glacier, Takeuchi et al. (2000)) and $z_0 = 0.016$ (Miage Glacier, Brock et al. (2010)). After a preliminary evaluation of our results and a comparison with the turbulent heat fluxes resulting from a point-scale energy balance at the location of the AWS on Piramide Glacier, we used a value of 6 mm for surface roughness.

In order to calculate G_{ratio} , it is necessary to estimate the vertical temperature profiles in the debris layer at the acquisition time of the satellite image (Equation A.3). In contrast to Rounce and McKinney (2014), we did not have buried thermistors to monitor debris temperature. As an alternative, we used the temperature profiles extracted from the point-scale surface energy balance at the location of the AWS on Piramide Glacier. This procedure allowed to follow one of the recommendations of Rounce and McKinney (2014), which is to use d_1 (Equation A.3) as thin as possible. We used d_1 as the debris thickness of the first layer in the point-scale energy balance model (2 cm).

Bands 10 and 11 of LANDSAT 8 satellite images are retrieved by the Thermal Infrared Sensor (TIRS) and are available at a nominal resolution of 30 m (100 m real resolution). Following the procedure described in the LANDSAT 8 user manual (USGS, 2015), we converted to top of atmosphere (TOA) radiance using Equation A.5 and then to at-

satellite brightness temperature using Equation A.6. We only used band 10 because band 11 was negatively affected by thermal energy from outside the field of view (i.e. stray-light) (USGS, 2015). The relevant equations are:

$$L_{\lambda} = ML \cdot Q_{cal} + AL \quad (A.6)$$

, where L_{λ} is the TOA spectral radiance [$W \cdot m^{-2} \cdot sr \cdot \mu m$], ML is Band-specific multiplicative rescaling factor from the metadata, AL is Band-specific additive rescaling factor from the metadata, Q_{cal} is quantized and calibrated standard product pixel values (DN).

$$T = \frac{K_2}{\log\left(\frac{K_1}{L_{\lambda}} + 1\right)} \quad (A.7)$$

, where T is at-satellite brightness temperature [K] and K_1 and K_2 are band-specific thermal conversion constants from the metadata.

All input data and the surface temperature map obtained as described in Eq. A.6 to A.7 were used to solve Eq. A.5 and to derive a map of debris thickness that we assume to be constant over the period of simulations.

Table 1: Main characteristics of Bello, Yeso and Piramide glaciers

Glacier	Area [km ²]	Percentage of debris covered area [%]	Elevation range [m asl]	Mean elevation [m asl]	Glacier length [km]
Bello	4.1	2.9	3989-4925	4430	3.9
Yeso	2.2	4.1	3842-5181	4444	3.7
Piramide	4.4	100	3230-4593	3670	7.5

Table 2: Summary of AWSs

AWS Name	Location	Coordinates (E,N) [UTM 19S]	Elevation [m asl]	Time periods	AWS	Measured variable (sensor)
BE	Bello Glacier	412744 6289113	4134	1.- 14 November 2013 to 15 April 2014	1	Wind speed and direction (Young 05103-5)
				2.- 1 October 2014 to end of study period		
YE	Yeso Glacier	414554 6289714	4428	There is a data gap between 20 January and 6 February 2013 due to a problem with the battery supply.	2	Incoming and outgoing shortwave and longwave radiation (Kipp & Zonen CNR4)
				1.- 15 November 2013 to 15 April 2014		
PI13	Piramide Glacier	417222 6285012	3655	2.- 5 November 2014 to end of study period	3	Air temperature and relative humidity (Young 41382)
				The station had a large datagap between 27 December 2014 and 24 February 2015, as the AWS fell into a crevasse and the battery was lost.		
PI14	Piramide Glacier	417353 6282990	3494	1.- 6 November 2013 to 15 April 2014	4	Incoming and outgoing shortwave and longwave radiation (Kipp & Zonen CNR2)
YE-off	Next to Yeso Glacier	414304 6288762	4300	1.- 15 November 2013 to 5 November 2014		
PI-off	Next to Piramide Glacier	415828 6280577	3022	1.- 5 November 2014 to end of study period		Air temperature and relative humidity (HMP110)

Table 3: Weather stations from the DGA network used in this study

Name	Coordinates (Lat,Lon)	Elevation [m asl]
Embalse El Yeso (Base station for precipitation)	33.677S, 70.089W	2475
Laguna Negra (Base station for air temperature)	33.666S 70.108W	2780
Aguas Panimávida	33.806S 70.070W	2237

Table 4: TOPKAPI-ETH calibrated parameters. The most sensitive parameters are in bold and a plausible range is given in parentheses. The plausible range is defined by the values that produce variations of $\pm 5\%$ in the total annual runoff of year 2014-15.

Name	Description	Calibrated value (plausible range)	Units	Sensitivity of total annual runoff to parameter variations	Units
<i>Air temperature distribution (Step 1)</i>					
<i>T_{grad}</i>	Monthly average temperature gradients	Jan = -7.06 (-6.8 – -7.3) Feb = -6.44 (-6.2 – -6.7) Mar = -5.94 (-5.7 – -6.2) Apr = -5.56 (-5.4 – -5.8) May = -5.30 (-5.1 – -5.5) Jun = -5.16 (5.0 – -5.4) Jul = -5.14 (-5.0 – -5.3) Aug = -5.24 (-5.1 – -5.4) Sep = -5.46 (-5.3 – -5.7) Oct = -5.80 (-5.6 – -6.0) Nov = -6.26 (-6.0 – -6.5) Dec = -6.84 (-6.6 – -7.1)	°C/km	-13.8	mm % ⁻¹
<i>T_{mod}</i>	Temperature decrease over snow and bare ice	1	°C	-0.2	mm % ⁻¹
<i>T_{mod_{debris}}</i>	Temperature increase over glacier debris	0.3	°C	0.1	mm % ⁻¹
<i>Snow and ice melt (Step 2)</i>					

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SRF	Parameters of ETI melt model	0.0048 (0.0041 – 0.0055)	$\text{mm m}^2 \text{h}^{-1} \text{W}^{-2}$	3.5	$\text{mm \%}^{-1}$
TF		0.1049	$\text{mm h}^{-1} \text{°C}$	0.4	$\text{mm \%}^{-1}$
T_T		-1 (-3.5 – 0.6)	°C	-22.8	mm °C^{-1}
TF_{d1}		0.0170		0.8	$\text{mm \%}^{-1}$
TF_{d2}	Parameters of D-ETI melt model	0.4043		0.8	$\text{mm \%}^{-1}$
SRF_{d1}		0.0114		0.8	$\text{mm \%}^{-1}$
SRF_{d2}		44.2		-0.9	$\text{mm \%}^{-1}$
lag_1		36			
lag_2		1			
α_{max}	Albedo of fresh snow	0.83 (0.72 – 0.91)		-4.6	$\text{mm \%}^{-1}$
r_d	Decay of snow albedo	0.102		0.9	$\text{mm \%}^{-1}$
<i>Precipitation distribution (Step 3)</i>					
P_{grad} 2013-14	Logarithmic coefficients for precipitation gradient	-5	-		
P_{grad} 2014-15	Logarithmic coefficients for precipitation	-5 (-4.97 - -5.10)	-	-40.7	$\text{mm \%}^{-1}$

	gradient	-43.1 (-43.19 - -43.28)		41.8	
P_T	Air temperature threshold to distinguish between solid and liquid precipitation	1 (-1.5 – 3.5)	°C	-2.3	mm °C ⁻¹
<i>Glacial meltwater routing(step 5)</i>					
k_{ice}	Storage constant for ice melt	2.4	Hours	0	mm % ⁻¹
k_{snow}	Storage constant for snowmelt	10.7	Hours	0	mm % ⁻¹
<i>Snow redistribution by gravity (Ragettli et al., 2015)</i>					
SGR_a	Exponential regression function	0.17245	-	-0.9	mm % ⁻¹
SGR_c	Snow holding depth dependent on the slope angle	250	m	0.1	mm % ⁻¹
<i>Melt delay at the beginning of the season (Fatichi et al., 2015; Ragettli et al., 2015)</i>					
MeltOnsetTimesteps	Melt delay at the beginning of the season	168	Hours		
MeltOnsetTemp	Threshold temperature for melt delay at the beginning of the season	0	°C		

Table 5: Simulated glacier mass balances and ELAs

Glacier	Total mass balance with avalanching[m]		Total mass balance without avalanching[m]		ELA [m asl]	
	2013-14	2014-15	2013-14	2014-15	2013-14	2014-15
Bello	0.02	-1.86	-0.27	-1.96	4419	4775
Yeso	-0.01	-1.79	-0.19	-1.83	4469	4848
Piramide	-0.55	-1.70	-1.35	-1.93	3881	4050

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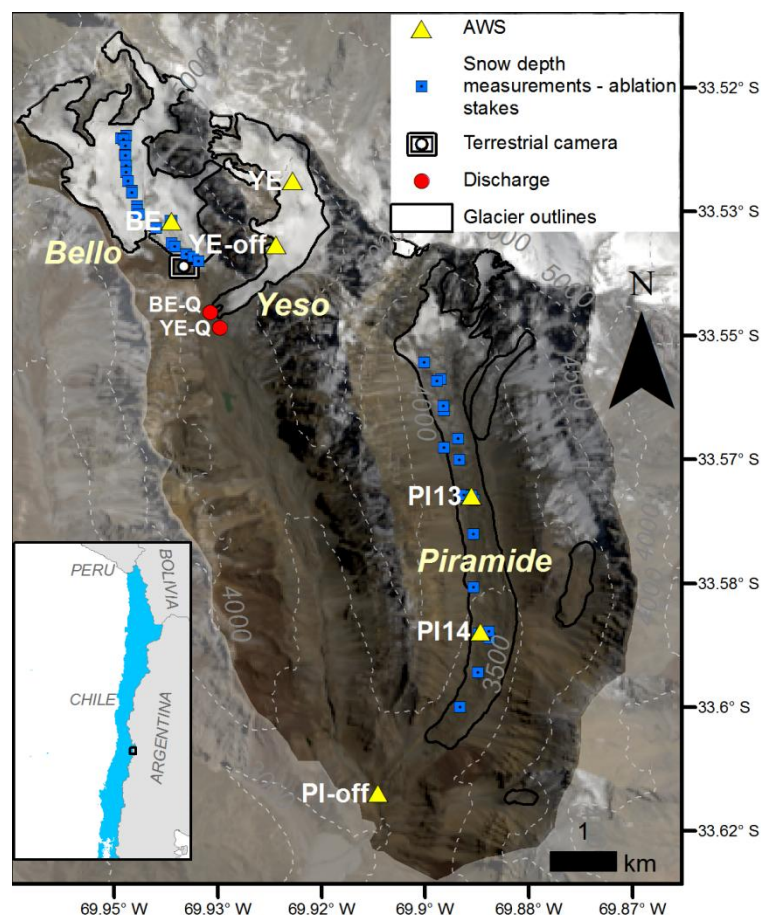


Figure 1: Location of Río del Yeso catchment in the semi-arid Andes of central Chile (2972-5476 m asl). We show the location of AWSs (BE, YE, YE-off, PI13, PI14 and PI-off), the terrestrial camera, the manual measurements of streamflow (BE-Q and YE-Q) and snow depth and ablation stakes (blue squares). Background image corresponds to a true colour Landsat 8 retrieved on 19 February 2014.

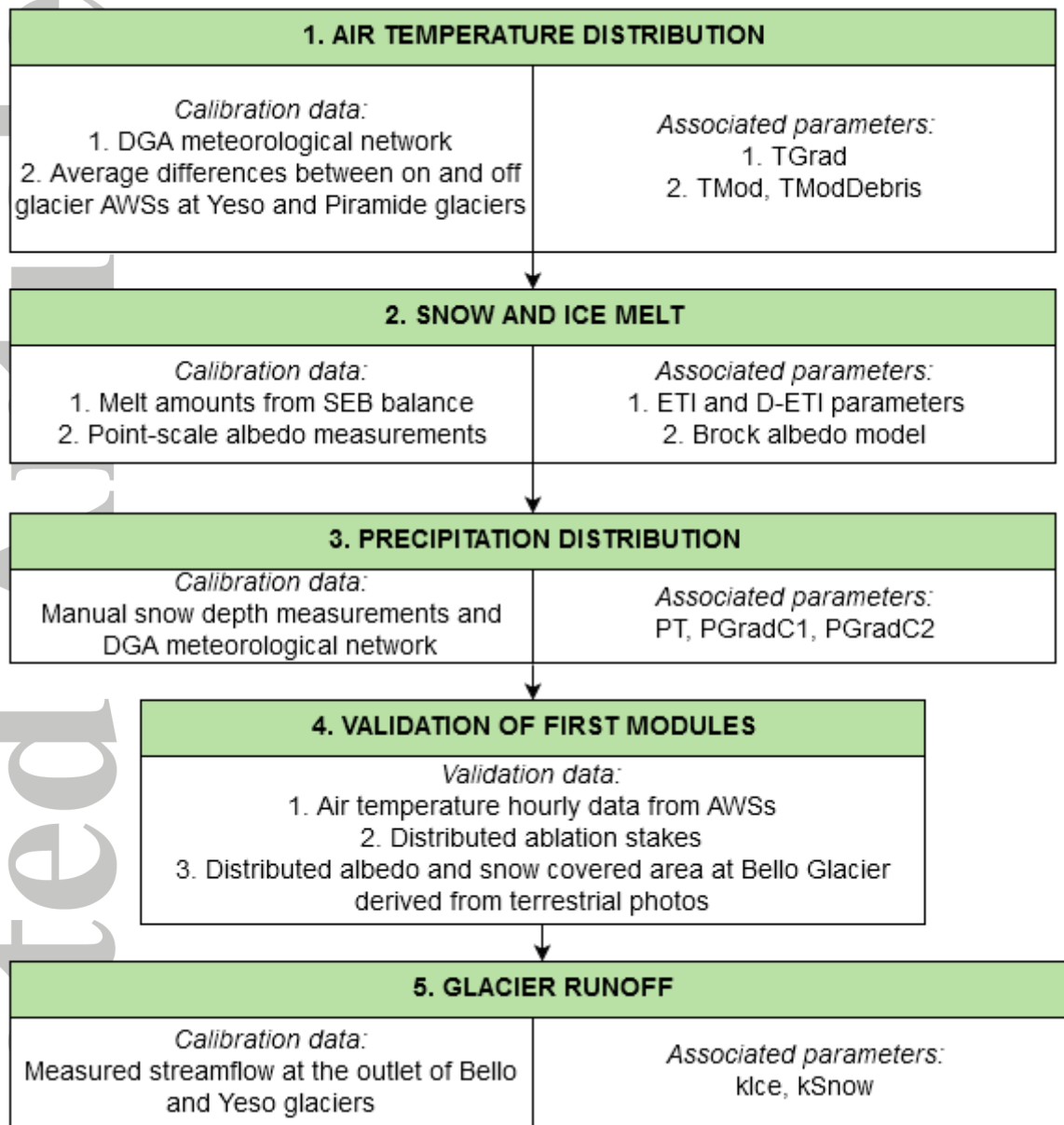


Figure 2: Strategy followed to calibrate the parameters values of TOPKAPI-ETH and to validate its results. Each step of this methodology has a calibration or validation dataset and, in the case of calibration steps, a set of associated TOPKAPI-ETH parameters.

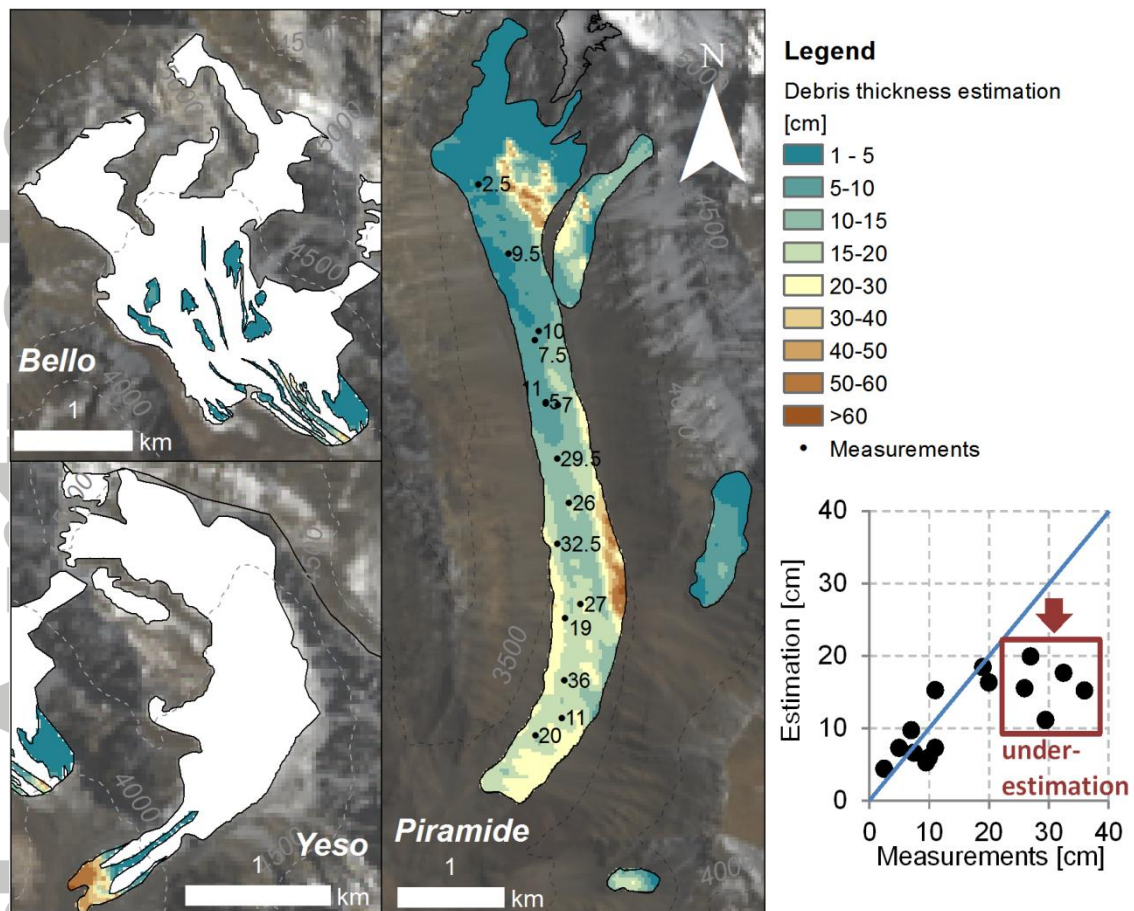


Figure 3: Debris cover thickness map derived using Rounce and McKinney (2014) method from the LANDSAT 8 image for Bello, Yeso and Piramide glaciers and three rock glaciers to the east of Piramide Glacier.

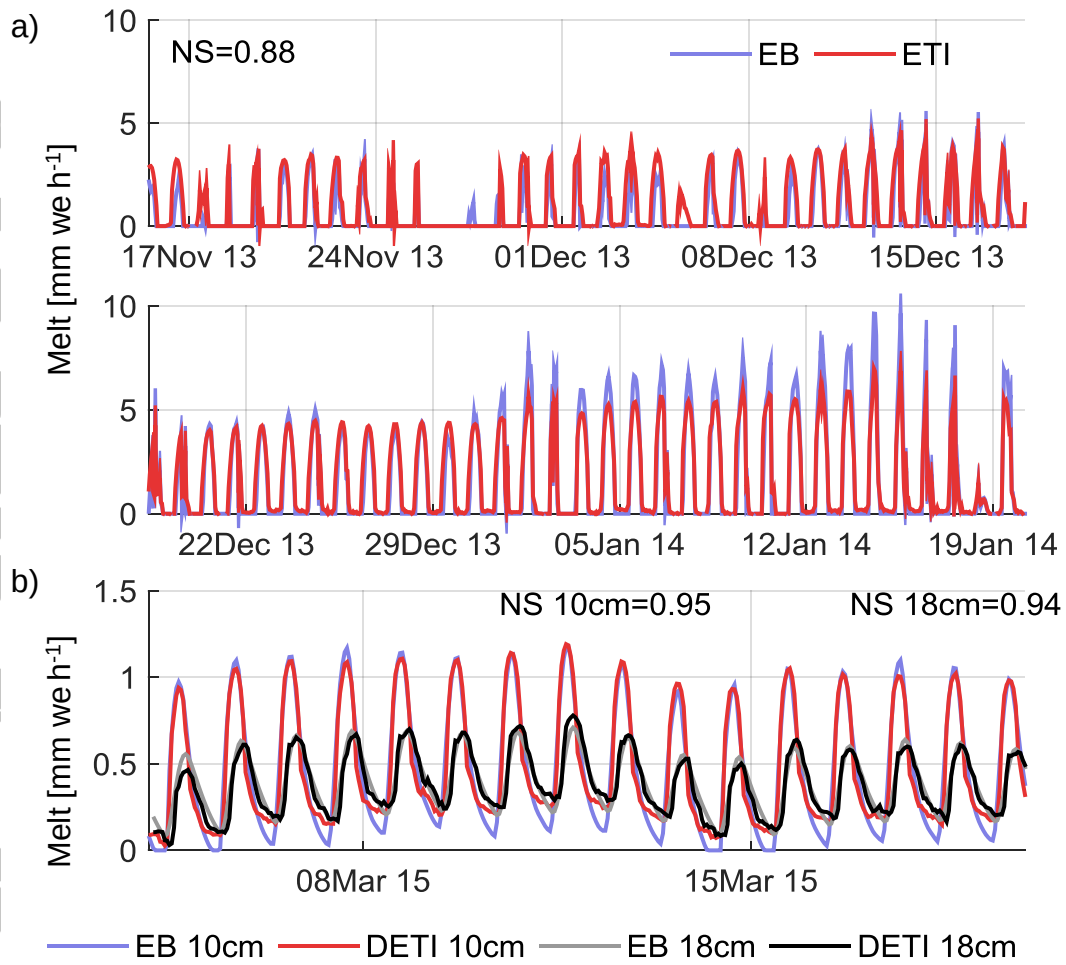


Figure 4: Calibration of melt parameters (Step 2, Figure 2). (a) Calibration of the ETI model using hourly melt rates from an energy balance model as reference. The calibration procedure was performed using data from the AWS on Bello Glacier during the ablation season 2013-14. We obtain a Nash-Sutcliffe coefficient of 0.88. (b) Calibration of the D-ETI model for different debris thicknesses. Reference melt amounts come from an energy balance model (EB) run with data from the AWS on Piramide Glacier during the ablation season 2014-15. The site has a debris thickness of 18 cm. We obtain Nash-Sutcliffe values of 0.95 and 0.94 for a debris thickness of 10 and 18 cm, respectively.

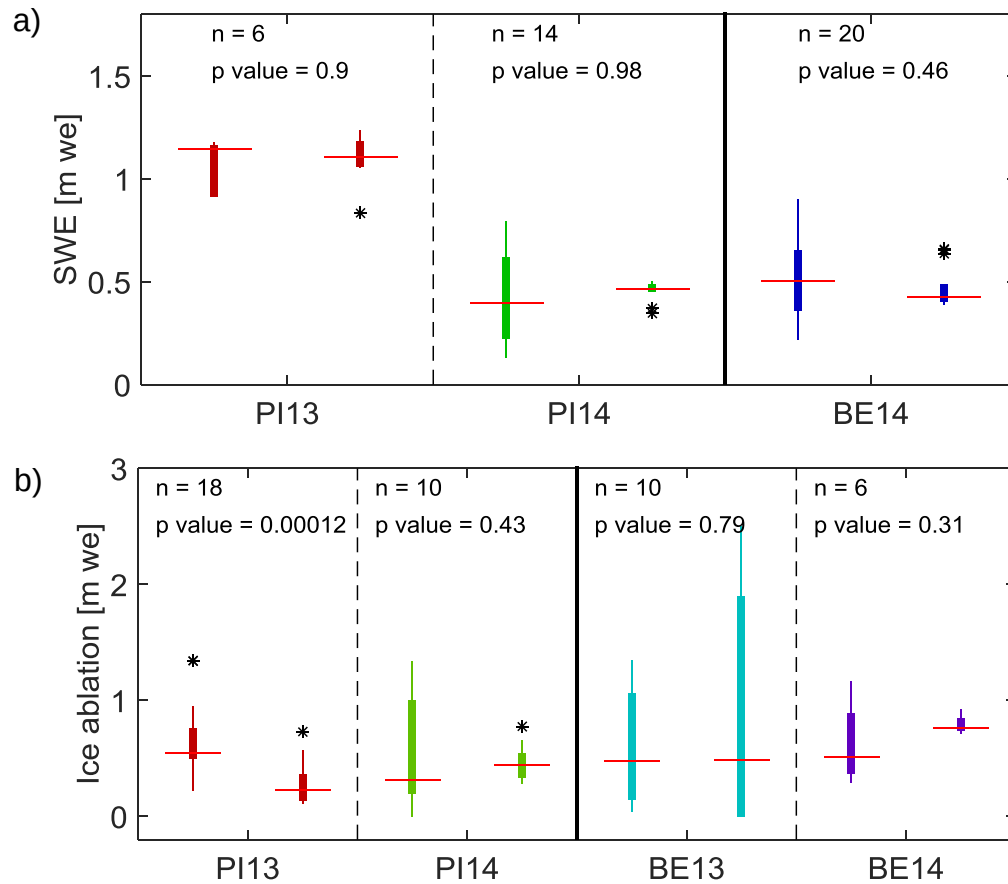


Figure 5: (a): Calibration of precipitation gradients (Step 2, Figure 2). TOPKAPI-ETH simulated values of snow depth compared to observed values at Piramide (PI) and Bello (BE) glaciers during 2013 and 2014. (b) Validation of ice ablation amounts (Step 4, Figure 2). TOPKAPI-ETH simulated values of ice ablation compared to observed values along Piramide (PI) and Bello (BE) glaciers during 2013 and 2014. At each site and season, left and right boxplots represent observed and simulated values, respectively. In both plots, medians are shown as horizontal red lines and outliers as black asterisks. We also show the number of observations (n) used to produce the boxplots and the p-value of the Wilcoxon non-parametric test for equal medians with a significance level of 5%. The test cannot reject the null hypothesis of equal medians for all the shown comparisons except for the ice ablation at Piramide 2013 (PI13).

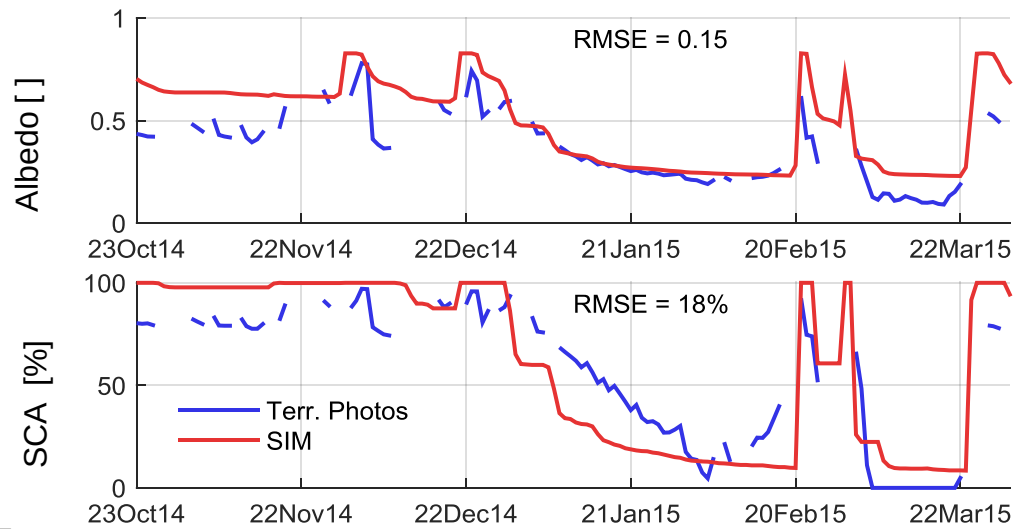


Figure 6: Validation of distributed surface albedo (Step 4, Figure 2). Comparison of observed and simulated time series of mean albedo and snow covered area over the area monitored using the terrestrial camera at Bello Glacier. Albedo was spatially averaged over the area observed by the camera. A threshold of 0.3 was used to distinguish between snow-covered and snow-free grid cells. RMSE values are indicated in the plots.

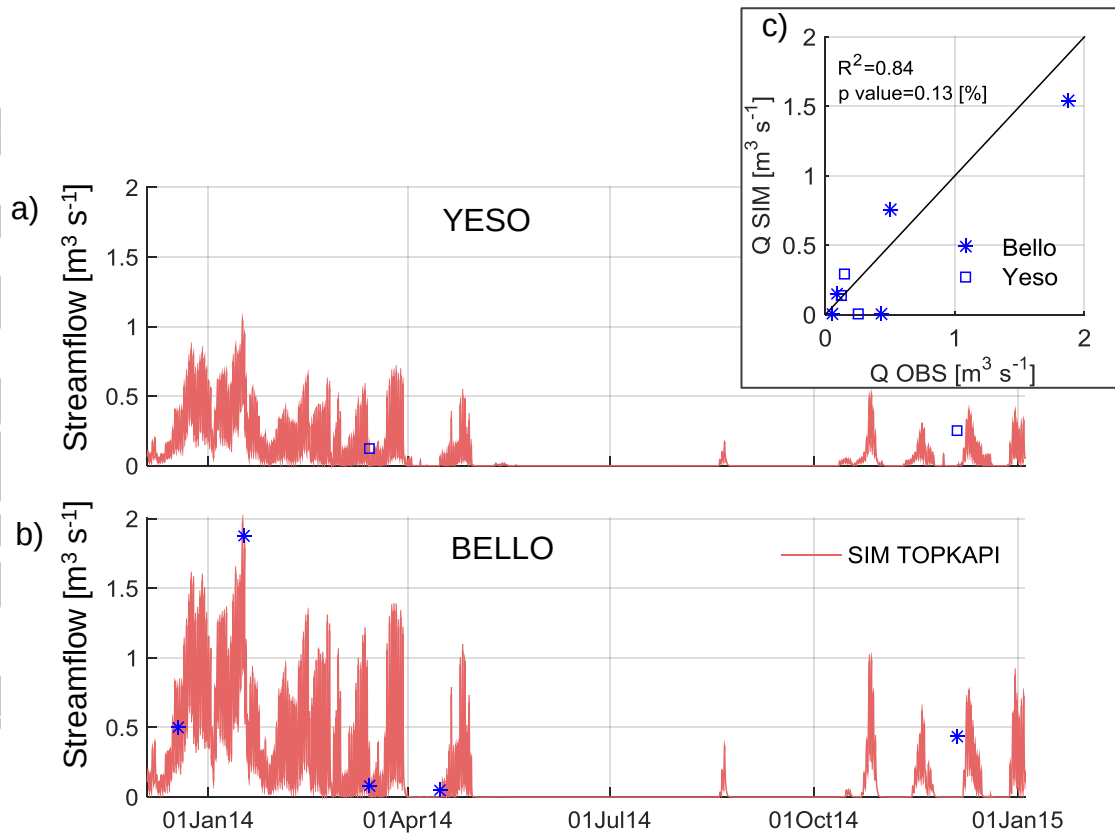


Figure 7: Calibration of the water storage coefficients of snow and glacial ice (Step 5, Figure 2). (a-b) TOPKAPI-ETH simulated values of glacier streamflow were compared to individual observations at the outlet of Yeso and Bello glaciers. (c) Scatter plot of simulated and observed values. The p value of the R^2 coefficient corresponds to the probability of getting the observed correlation by random chance, when the true correlation is zero. Note that the last observation at Bello and Yeso glaciers was not captured by the model, probably due to lateral or groundwater runoff contributions.

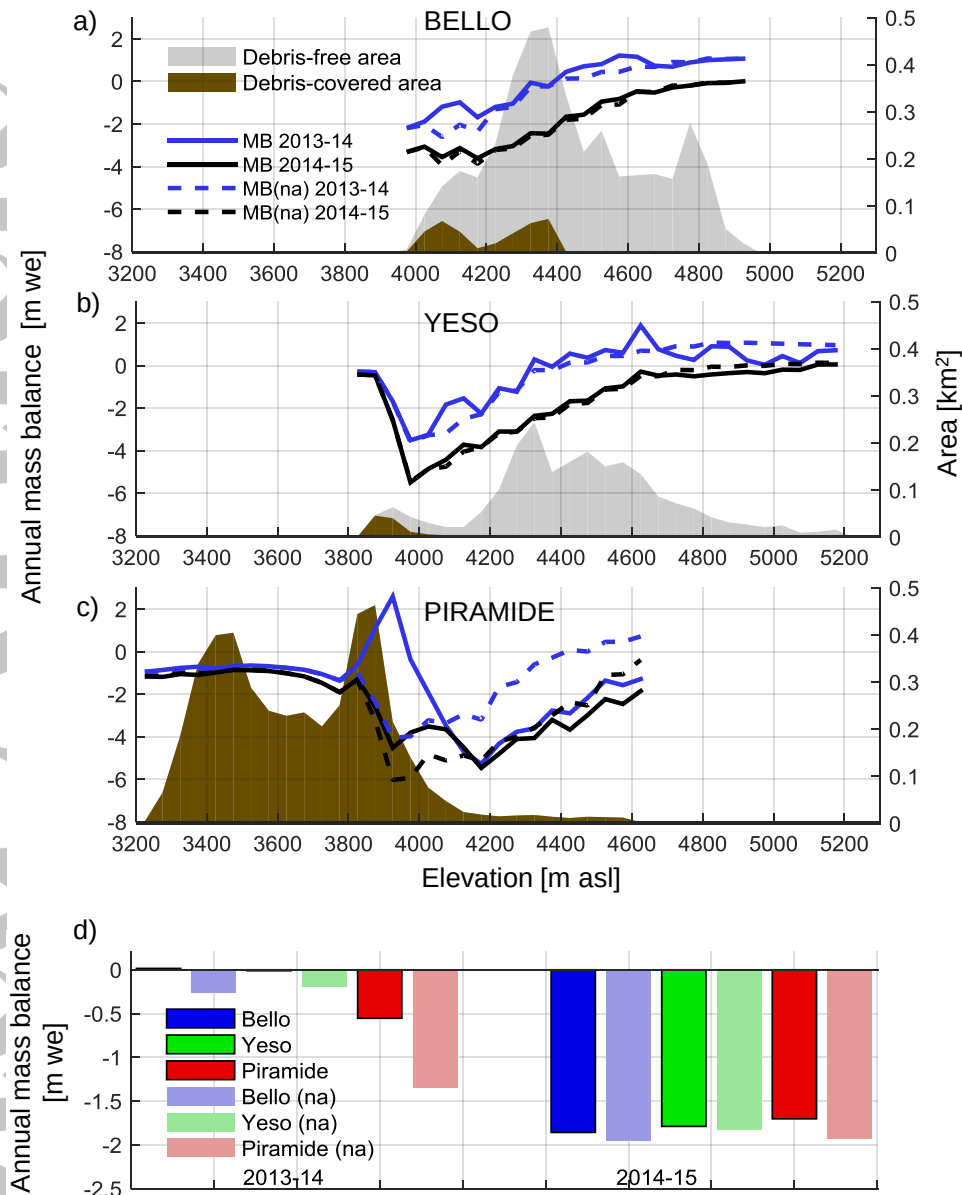


Figure 8: (a) to (c) Simulated annual glacier mass balance of both years as a function of elevation and debris-free and debris-covered areas in Bello, Yeso and Piramide glaciers. Bold and dashed lines represent the mass balance resulting from simulations with and without the snow gravitational redistribution module, respectively. (d) Total glacier mass balance for both analysed seasons. At each glacier, the left (dark colour) and right bars (light colour) represent the mass balance resulting from simulations with and without the snow gravitational redistribution module, respectively.

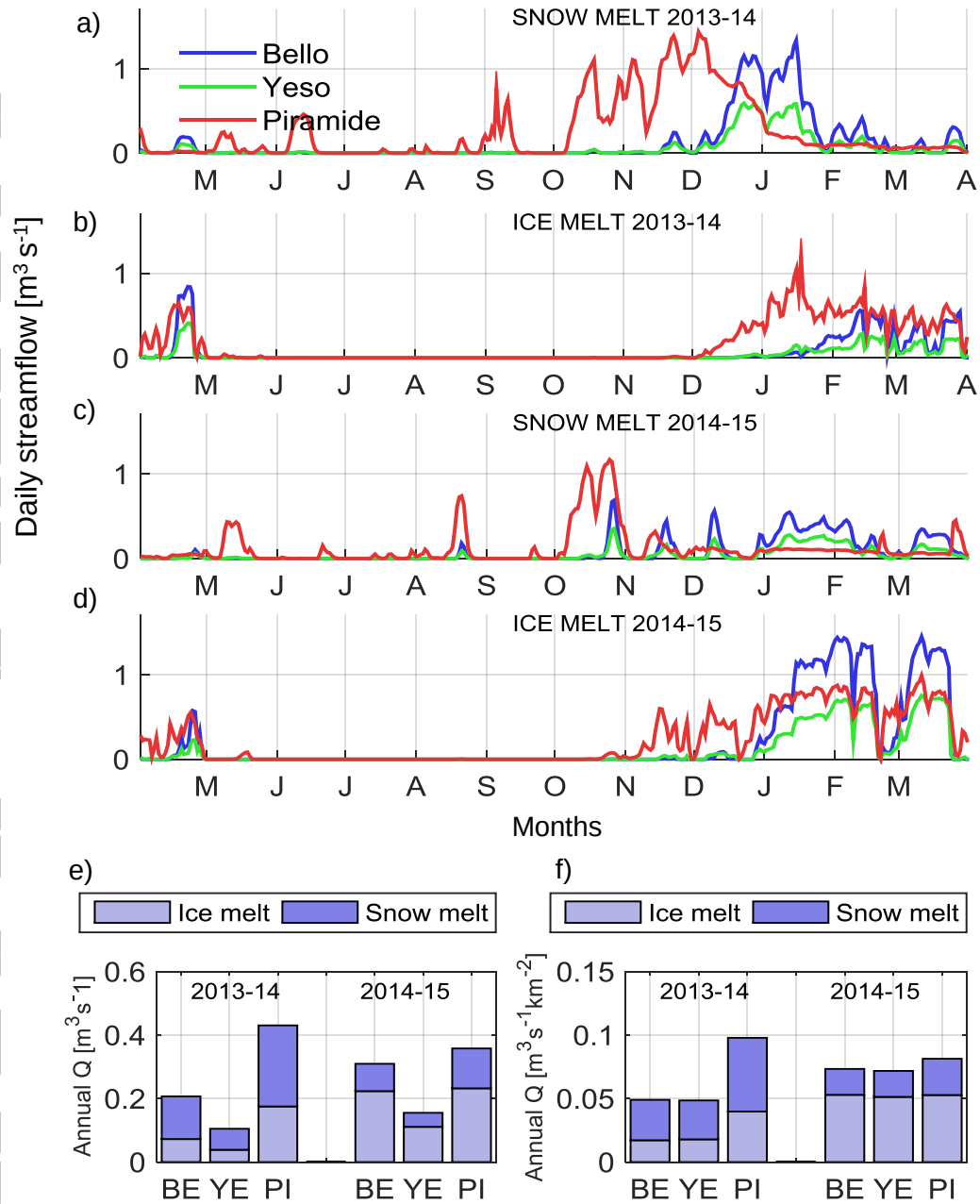


Figure 9: (a-e) Daily mean streamflow from snow and ice melt simulated by TOPKAPI-ETH for Bello, Yeso and Piramide glaciers in the two analysed years. (e) and (f) show an annual summary of these results for runoff and specific runoff (normalized by the glacier areas).

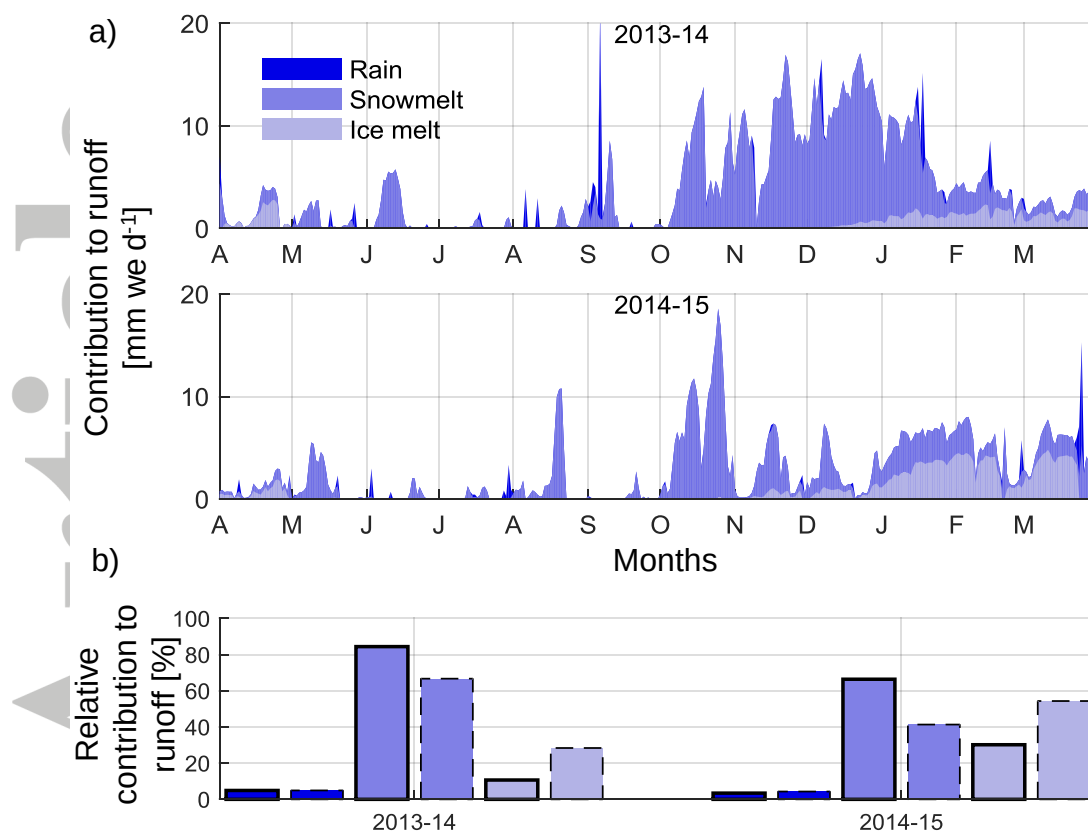


Figure 10: Summary of contributions to runoff simulated by TOPKAPI-ETH. a) Daily total contribution of rain, snowmelt and ice melt to the catchment runoff during years 2013-14 and 2014-15. (b) Annual and summer relative contributions of rain, snowmelt and ice melt. At each runoff component left (bold edges) and right bars (dashed edges) represents annual and summer contributions.